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/ **Practice Exam - 1**

Correct Answer

Next C

Practice Exam - 1

Question 49 of 117



A 43-year old female patient is status post heart transplantation 2 months ago. Her post-transplant course was notable for treatment of fungal infection with fluconazole for 6 weeks but was otherwise without incident. Her immunosuppression regimen consists of prednisone, tacrolimus and mycophenolate mofetil. Her tacrolimus trough level was maintained in the 10-12 ng/mL range. She now presents with dyspnea, edema and is found to have a newly depressed ejection fraction of 45%. The patient reports a gastrointestinal illness characterized by 2 days of profuse diarrhea which self-resolved last week. She was able to take her immunosuppressive medications during the episode of diarrhea. Her current tacrolimus trough is 4.4 ng/mL.

Which of the following is most likely to decrease tacrolimus levels?



A. Discontinuation of fluconazole



B. Recent diarrheal illness

C. Nonadherence to immunosuppressive medications

D. Inappropriate timing of tacrolimus level assessment

Commentary

References

Testing Point: The Board-certified AHFTC specialist should recognize drug-drug interactions when managing immunosuppression regimens post-transplantation.

Answer: A. Discontinuation of fluconazole.

Tacrolimus is a calcineurin inhibitor (CNI) and part of the backbone of the immunosuppression regimen post transplantation. Several drug-drug interactions may occur and should be considered when aiming for optimal drug level maintenance to minimize risk of infection and rejection. The azole antifungal agents inhibit metabolism of CNI through their interactions with CYP3A4 and P-glycoprotein, both of which are important in CNI metabolism. Thus, patients on fluconazole will often need lower doses of tacrolimus to maintain adequate serum levels. When the patient was taken off fluconazole two weeks prior to presentation, this likely resulted in

reduced tacrolimus levels due to less inhibition of CYP3A4 and P-glycoprotein by the azole and increased metabolism of tacrolimus.

Answer B: P-glycoprotein is a membrane associated protein. In the intestine, it causes intestinal efflux of drug from the enterocyte back into the gut, thereby reducing CNI levels. However, in the setting of severe diarrhea, intestinal epithelial cells may actually be destroyed, thereby abrogating P-glycoproteins and increasing CNI levels through increased absorption. Therefore, answer B is incorrect, as tacrolimus levels would have been expected to have increased during the patient's recent severe diarrheal illness.

Though nonadherence to immunosuppression medications (answer C) can result in decreased tacrolimus levels, the discontinuation of the azole is more likely contributing. Drug adherence and tolerance should be routinely assessed post-transplant.

Answer D: Inappropriate timing of tacrolimus levels would result in a higher measured tacrolimus level if drawn soon after a dose is given, or prior to the actual trough.

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